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MILITARY SPECIFICATION

BATTERY, SILVER-ZINC-ALKALI GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the design, construction, inspection, and preparation for delivery of a silver-zinc-alkali battery.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bid or request for proposal form a part of this specification to the extent specified herein.

* SPECIFICATIONS

Military

MIL-P-116 - Preservation, Methods of

MIL-E-5272 - Environmental Testing, Aeronautical and Associated Equipment, General Specifications For

MIL-T-7928 - Terminals, Lug and Splice, Crimp Style, Copper

Beneficial comments (recommendations additions, deletions) any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC:6140

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MIL-E-16400 - Electronic Equipment, Naval Ship and Shore, General Specification

MIL-W-16878 - Wire, Electrical, Insulated, High Temperature (Navy)

MIL-P-19834 - Plates, Identification, Metal Foil, Adhesive Backed

MIL-F-22191 - Films, Transparent, Flexible, Sealable, for Packaging Application

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-130 Identification Marking of U.S. Military Property

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts

MIL-STD-454 - Standard General Requirements for Electronic Equipment

DRAWINGS

Naval Air Systems Command

DL-632AS100 - Battery, Silver-Zinc-Alkali

67A76F1 - Battery Assembly, Silver-Zinc-Alkali, for AQM-37A Missile Target; Interface Control Drawing

Naval Ordnance Systems Command

2580161 - Container, Shipping and Storage (Battery and Filling Kit)

PUBLICATIONS

Naval Ordnance Systems Command

OP-2230 - Workmanship and Design Practices for Electronic Equipment

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bid or request for proposal shall apply.

Department of Transportation

DOT-2E (49 CFR 178.24a) and Appendix B of 49 CFR 178 (Plastic Polyethylene)

National Bureau of Standards

FED-STD-H28 - Screw Thread Standards for Federal Services

(Applications for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS

3.1 Description. The battery covered by this specification consists of a battery assembly, filling kit, electrolyte, and instructions.

3.1.1 Battery Assembly. The battery assembly, shown on Drawing 67A76F1, consists of a battery unit and a box assembly.

3.1.1.1 Battery Unit. The battery unit consists of a main battery of 19 silver-zinc-alkali cells and a cartridge battery of 6 similar cells, each equipped with a removable vent plug. The battery unit includes a spacer block, a heater, a thermostatic switch, the cell interconnectors, a wiring harness, and a connector receptacle.

3.1.1.2 Box Assembly. The box assembly encases the battery unit and consists of a box with a hinged lid and the lid fasteners.

3.1.2 Battery Accessories. Battery accessories consist of a filling kit and the screws for installing the battery in the AQM-37A Missile Target.

3.1.2.1 Filling Kit. The filling kit consists of vent cleaners, tweezers, absorbent cotton, filling caps, and extra vent plugs, all in sufficient quantities to permit compliance with the instructions furnished by the contractor (see 3.1.4).

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3.1.2.2 Installation Screws. Screws for installing the battery in the target are those designated as item 4 on Drawing 67A76F1.

3.1.3 Electrolyte. The electrolyte shall be furnished in sufficient quantity to fill 27 cells.

* 3.1.4 Instructions. A set of service and operating instructions shall be provided with each battery. The instructions shall include safety procedures and precautions, preparation for use, storage conditions, recharging, and maintenance.

3.2 General Requirements.

3.2.1 First Article Sample. Five first article samples of the battery shall be manufactured using the methods and procedures proposed for the production lot. These samples shall be tested as specified in Section 4 to determine that the first article item will satisfy this specification. All components used in the production items shall be identical in every respect (methods, procedures, manufacturer, etc.) to those used in the accepted first article sample.

3.2.2 Materials. All materials entering into the final assembly of the battery shall conform to the applicable Federal or Military specification. Materials which are not covered by specification or which are not specifically designated herein shall be of the best quality, of the lightest practicable weight consistent with strength and durability, and entirely suited to the purpose for which intended under all conditions of test and service operation.

3.2.2.1 Magnesium. The use of magnesium requires the specific approval of NAVAIR or its appointed agent.

3.2.2.2 Flammable Materials. Materials which will support combustion or which are capable of causing an explosion shall not be used.

3.2.2.3 Toxic Materials. Materials, exclusive of the electrolyte, which may produce toxic or corrosive effects shall not be used.

3.2.2.4 Dissimilar Metals. The use of dissimilar metals shall be in accordance with MIL-E-16400.

3.2.3 Moisture, Fungus, and Corrosion. The battery shall be moisture, fungus, and corrosion resistant as specified in MIL-E-16400.

3.2.4 Protective Coatings and Surface Treatments. The protective coatings and surface treatments used for the protection of the battery assembly shall be suitable for the electrolyte used and for the environmental conditions to which the battery assembly will be subjected. Materials and protective treatments used shall be satisfactory for corrosion and for stress cracking behavior.

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3.2.5 Marking. Parts and assemblies shall be marked in accordance with MIL-STD-130.

3.2.5.1 Nameplates. Nameplates shall conform to MIL-P-19834, Type I.

3.2.5.2 Terminal Markings. The polarity of the battery terminals shall be permanently marked on the end of each battery cell adjacent to the terminals.

3.2.5.3 Additional Information. Each battery shall bear the date of manufacture and a serial number which will be assigned by the Government Inspector upon request, or as specified in the contract or purchase order.

3.3 Construction and Dimensions.

3.3.1 Production Drawings. The battery shall be fabricated and assembled in accordance with the drawings listed on DL 632AS100 and the documents specified herein.

3.3.1.1 Dimensions. Dimensions and tolerances for the finished battery assembly shall be as shown on Drawing 67A76F1.

3.3.1.2 Battery Unit Spacer Block. The battery unit spacer block shall have the external dimensions of a single cell less the filling hole, vent, and terminals. The block shall be cut away to provide a recess for the thermostatic switch and allow passage of the wire leads to the heater.

3.3.1.3 Cell Filling Hole. The cell filling hole shall be of sufficient size to allow insertion of the filler spout furnished as part of the filling kit.

3.3.2 Interchangeability. Components of the battery assembly shall be interchangeable and shall assemble with mating components of the battery assembly without selection or fitting.

3.3.3 Threads. All threads shall be in accordance with FED-STD-H28. The NOT-GO gaging practice shall be as set forth therein.

3.3.4 Wiring. The battery unit wiring shall conform to Figure 1.

3.3.4.1 Wire. Hook-up wire for internal wiring shall conform to MIL- W-16878, except that the wire shall have extruded insulation in lieu of braided insulation.

3.3.4.2 Terminal Lugs. Solderless or crimp-on terminal lugs shall be used for electrical connections to screw type terminals. The lugs shall conform to MIL-T-7928.

3.3.4.3 Pressure Contacts. Electrical connections between terminals shall be rigid metallic members.

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3.3.4.4 Cabling of Leads. Wires shall be arranged to permit grouping into prefabricated cables. Cables shall be laced with acceptable cord and secured by methods described in OP 2230.

3.3.4.5 Restriction. No more than three wires shall terminate at any one terminal.

3.4 Environmental Requirements.

* 3.4.1 Temperature. Altitude. When tested at a condition of 25,000 feet altitude and at temperatures of -65 degrees F and +125 degrees F the battery assembly shall show no damage or leakage of electrolyte. The average voltage of the main and cartridge batteries shall be not less than 26.8 and 8.46 volts direct current (Vdc) respectively at both temperature extremes. Except for five milliseconds after initial load application, the main battery voltage output shall not be less than 23.0 Vdc during the first 10 minutes of continuous 15 ampere rate discharge.

* 3.4.2 Temperature. Vibration. When subjected to the cycling vibration test of 20 to 1200 to 20 cycles per second at temperatures of -65 degrees F and +125 degrees F the battery assembly shall show no damage or leakage of electrolyte. The voltage of the main and cartridge batteries shall not fall below 25.0 amd 7.9 Vdc, respectively, at both temperature extremes.

3.4.3 Shipboard Vibration. When vibration tested in accordance with 4.4.3.4, the battery assembly shall show no damage or leakage of electrolyte.

* 3.4.4 Shock. When shock tested in accordance with MIL-E-5272, Procedure V, the battery assembly shall show no damage or leakage of electrolyte. The voltage of the main and cartridge batteries shall be not less than 26.8 and 8.46 Vdc, respectively.

* 3.4.5 Fungus. When fungus tested in accordance with MIL-E-5272, Procedure I, assembly shall show no damage and shall meet the requirements of 3.5.11.

3.5 Performance Requirements and Product Characteristics. The battery shall meet the following performance requirements and product characteristics.

3.5.1 Weight. The weight of the battery assembly shall be in accordance with Drawing 67A76F1.

3.5.2 Pressure Tightness. Each cell shall maintain an internal pressure of not less than 1.5 pounds per square inch gage (psig).

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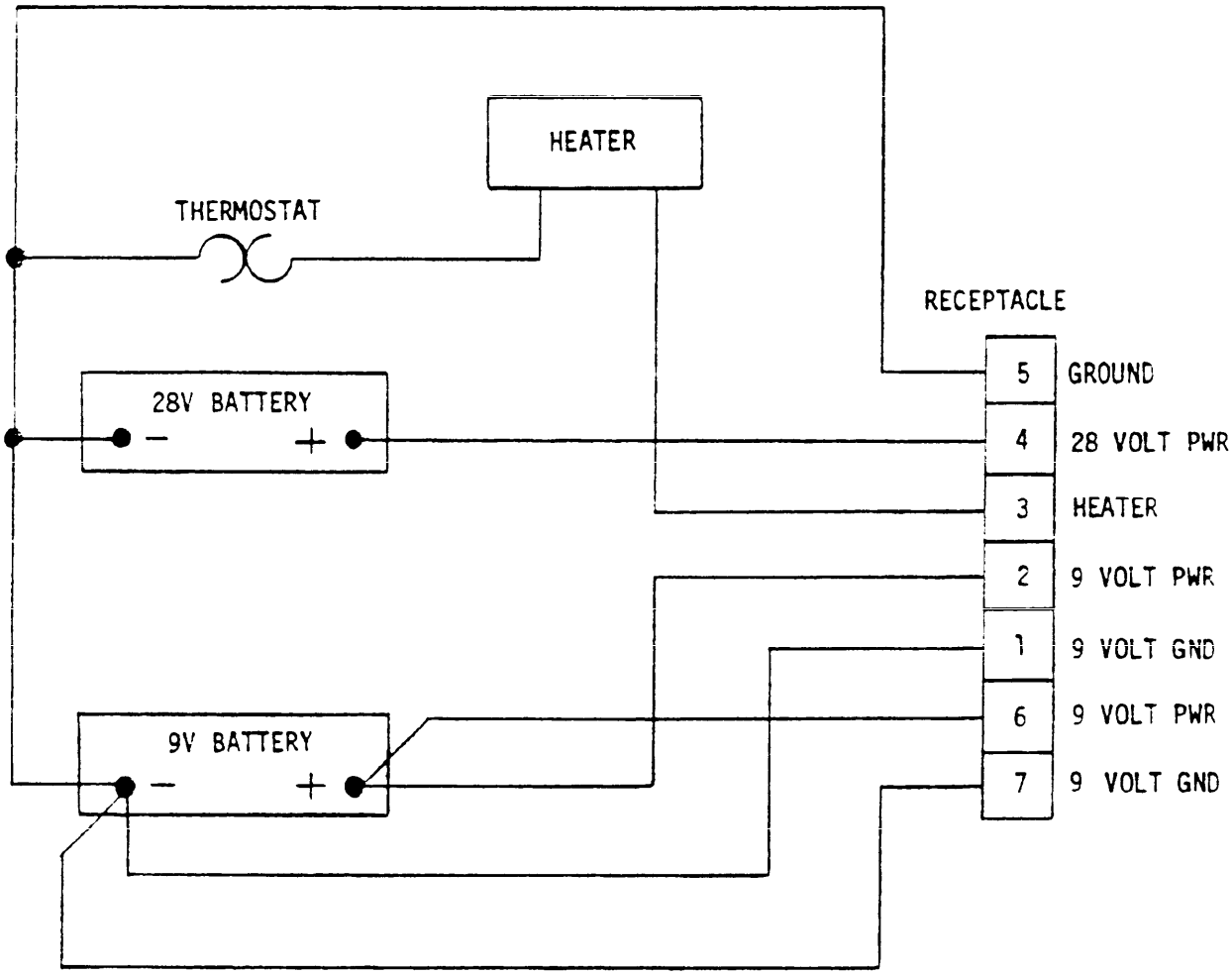


FIGURE 1. SCHEMATIC - BATTERY UNIT

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storage and shall be subjected to vibration

* 3.5.3 Heater. The heater shall heat the battery cells to its optimum operating temperature of +70 degrees F within 25 minutes when the battery is subjected to a temperature environment of -65 degrees F. The heater shall operate from a nominal 28 Vdc (28.5 maximum; 24 minimum) and shall have an internal resistance of not more than 8.8 ohms.

3.5.4 Thermostatic Switch. The switch shall control the operation of the heater to maintain the battery operating temperature within the operating range of +60 degrees F to +160 degrees F. The switch shall be capable of withstanding 100,000 cycles when subjected to a nominal voltage of 28 Vdc and a current of 4 amperes. When the switch is tested in accordance with 4.5.4.5, the cut-out and cut-in temperatures shall have a tolerance of plus or minus 3 degrees Fahrenheit (F).

3.5.5 Resistance.

3.5.5.1 Harness, Connector, Receptacle, and Heater. The harness, connector, receptacle, and heater shall have an insulation resistance of not less then 100 megohms and shall withstand the dielectric test without insulation breakdown, arcing, or other damage.

3.5.5.2 Cell Resistance, Dry. Each cell of the battery unit in a dry condition shall have an internal resistance of not less than 100,000 ohms.

3.5.5.3 Cell Resistance, Wet. Each cell of the battery in a wet and charged condition shall have an internal resistance not greater than 0.02 ohm.

3.5.6 Inversion. The battery shall be capable of being held in any position for 30 minutes without loss of electrolyte.

3.5.7 Cell Voltage. The open circuit voltage of each cell after initial activation and after the first recharge shall be between 1.32 and 1.86 Vdc.

* 3.5.8 Capacity, Main Battery. The time required to reach 20.9 Vdc shall be not less than 25 minutes. The average voltage during the first 25 minutes of discharge shall be not less than 26.8 Vdc. Except for five milliseconds after initial load application, the output voltage shall be not less than 23.0 Vdc during the first 10 minutes of continuous 15 ampere rate discharge.

* 3.5.9 Capacity, Cartridge Battery. The time required to reach 6.6 Vdc shall be not less than 25 minutes. The average voltage during the first 25 minutes of discharge shall be not less than 8.46 Vdc.

* 3.5.10 Operation Life. The main and cartridge batteries shall show no damage or leakage of electrolyte and shall provide five operating cycles.

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3.5.11 Capacity, Main and Cartridge Batteries After 30-Day Activated Stand Time. When tested in accordance with 4.5.4.10, the main and cartridge batteries shall reach voltages of 20.9 and 6.6 Vdc in not less than 23.75 minutes. The average voltage during the first 23.75 minutes of discharge shall not be less than 26.2 Vdc for the main battery and 8.28 Vdc for the cartridge battery. Except for five milliseconds after initial load application, the main battery output voltage shall be not less than 23.0 Vdc during the first 10 minutes of continuous 15 ampere rate discharge.

3.5.12 Storage Life. The battery when packaged in accordance with 5.1.2.1 and stored for a minimum of two years shall meet the capacity and operating requirements of this specification. The battery when subjected to the tests of 4.4.3.1 and 4.4.3.7 shall meet the requirements of 3.5.8, 3.5.9, and 3.5.10.

* 3.6 Cleanliness. Prior to and after final assembly, the battery unit shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, rosin flux, and other foreign materials.

* 3.7 Workmanship. The battery, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, and marking of parts and assemblies, and to plating, machine screw assembly, welding, and brazing, where applicable. The standards of workmanship exhibited in any approved preproduction sample shall be determinative of the requirements of the contract relative to workmanship. The workmanship standards of MIL-STD-454 shall be met.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or specify the place of performance of any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Inspections. Inspections of the battery shall be classified as follows:

a. First article inspection (4.4)

b. Quality conformance inspection (4.5)

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* 4.3 First Article Sample. A first article sample of five batteries shall be subjected to inspection at an activity designated by the procuring agency to determine conformance to this specification (see 6.2). Acceptance of the first article sample shall be based on no defects in the sample.

* 4.4 First Article Inspection. After the first article sample has passed the quality conformance examination and tests of 4.5.1, except the tests of 4.5.1.2(i), 4.5.1.2(l), 4.5.1.2(m), and 4.5.1.2(n) shall not be performed, the sample shall be subjected to the tests of Table I as detailed in 4.4.3.

4.4.1 Test Conditions. Except as modified under the specific tests of 4.4.3, ambient test conditions shall be as follows:

a. Temperature +65 degrees F to +95 degrees F

b. Humidity Up to 95% (relative)

c. Altitude Normal Ground

d. Vibration None

TABLE I. First Article Tests.

Sample Number	1	2	3	4	5	Requirement paragraph	Test method paragraph
Temperature, altitude	X	X				3.4.1	4.4.3.2
Temperature, vibration	X	X				3.4.2	4.4.3.3
Shipboard vibration			X			3.4.3	4.4.3.4
Wet cell resistance			X			3.5.5.3	4.5.4.6.3
Shock			X			3.4.4	4.4.3.5
Fungus				X		3.4.5	4.4.3.6
Storage life					X	3.5.12	4.4.3.1
Operating life					X	3.5.10	4.4.3.7

4.4.2 Test Equipment and Setup. The designated testing activity shall furnish and maintain all necessary test equipment, facilities, and personnel to perform all first article tests. The test equipment and facilities shall be adequate in quantity and, unless definite requirements are specified, shall be of sufficient accuracy and quality to permit performance of the required first article tests.

4.4.3 First Article Test Procedures.

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4.4.3.1 Accelerated Storage Life Test. The battery assembly without electrolyte and with battery heater not energized shall be subjected to the following test:

a. Place the battery assembly in a test chamber. The test chamber shall be vented to the atmosphere to prevent the buildup of pressure. Prior to starting the test period, the chamber temperature shall be plus (+) 68 degrees F to +100 degrees F with uncontrolled humidity.

b. Subject the battery assembly to a temperature of minus (-) 65 degrees F for 2 hours.

c. Change the temperature in the test chamber to +160 degrees F with relative humidity of 95% reached in 2 hours and maintained for 14 hours. Steam and distilled water having a pH value between 6.5 and 7.5 at +77 degrees F shall be used to obtain the desired humidity.

d. Decrease the temperature in the test chamber to -65 degrees F at the rate of 36 degrees F per hour. Maintain at -65 degrees F for 2 hours.

e. The cycle of paragraphs (c) and (d) shall be repeated 10 times. At the conclusion of the test period the battery assembly shall be returned to ambient conditions. Dry the battery assembly by wiping. Do not use an oven or air blast for drying.

f. Fill battery with electrolyte.

g. Examine battery for damage and leakage of electrolyte.

4.4.3.2 Temperature, Altitude. The battery assembly shall be activated (electrolyte added) and allowed to soak for one hour prior to starting the test.

a. The internal temperature of the test chamber shall be reduced to -65 degrees F and the battery inserted into the test chamber. There shall be no load on the battery and the battery heater shall be operating from external power. The pressure within the test chamber shall now be reduced to that which is encountered at an altitude of 25,000 feet. After a period of two hours under these conditions, the battery heater shall be turned OFF and

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a simulated target load of 15 amperes shall be placed on the main and cartridge batteries and maintained for 13 minutes. The main battery voltage shall be measured and recorded continuously for the first minute of discharge with a recorder capable of recording voltage during the first five milliseconds of discharge. The voltage of both the main and cartridge batteries shall then be measured and recorded at one minute intervals during battery discharge. The battery shall be examined for damage and leakage of electrolyte and then shall be recharged.

b. The internal temperature of the test chamber shall now be raised to +125 degrees F. There shall be no load on the battery and the battery heater shall not be operating. The pressure within the test chamber shall be reduced to that which is encountered at an altitude of 25,000 feet. After a period of two hours under these conditions, a simulated target load of 15 amperes shall be placed on the main and cartridge batteries and maintained for 13 minutes. The main battery voltage shall be measured and recorded continuously for the first minute of discharge with a recorder capable of recording voltage during the first five milliseconds of discharge. The voltage of both the main and cartridge batteries shall then be measured and recorded at one minute intervals during battery discharge. The battery shall be examined for damage and leakage of electrolyte and then shall be recharged.

4.4.3.3 Temperature, Vibration. The battery assembly shall be mounted in a device which simulates normal target mounting. The battery assembly shall be activated (electrolyte added) and placed in the test chamber with battery heater activated from an external power source.

a. The internal temperature of the test chamber shall be reduced to -65 degrees F. With no load on it, the battery assembly shall be subjected to a cycling vibration test of 13-minute sweeps of frequencies from 20 to 1200 to 20 cycles per second (cps) along each of three mutually perpendicular axes. The maximum force shall be two g. The variation of frequency shall be logarithmic. If any resonances are encountered, each resonance shall be maintained for five minutes. The total vibration time along each axis shall be two hours.

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b. The battery heaters shall now be turned OFF and a simulated target load of 15 amperes shall be placed on the main and cartridge batteries. The test chamber shall be maintained at an internal temperature of -65 degrees F. Simultaneously with the imposition of the battery load, the battery assembly shall be subjected to a cycling vibration test of one 13-minute sweep from 20 to 1200 to 20 cps along each of three mutually perpendicular axes. The maximum force shall be two g and the variation in frequency shall be logarithmic. Any resonant frequency previously determined shall be sustained for one minute. The voltage of the main and cartridge batteries shall be measured and recorded at one minute intervals during the discharge period. Between discharges and upon completion of the test, the battery assembly shall be recharged. Between discharges the battery heater may be activated at the discretion of the testing activity.

c. The internal temperature of the test chamber shall be raised to +125 degrees F. With no load on it, the battery assembly shall be subjected to a cycling vibration test of 13-minute sweeps of frequencies from 20 to 1200 to 20 cps along each of three mutually perpendicular axes. The maximum force shall be two g and the variation in frequency shall be logarithmic. If any resonances are encountered, each resonance shall be maintained for five minutes. The total vibration time along each axis shall be two hours.

d. A simulated target load of 15 amperes shall now be placed on the main and cartridge batteries. The test chamber shall be maintained at an internal temperature of -125 degrees F. Simultaneously with the imposition of the battery load, the battery assembly shall be subjected to cycling vibration test of one 13-minute sweep from 20 to 1200 to 20 cps along each of the three axes. The maximum force shall be two g and the variation in frequency shall be logarithmic. Any resonant frequency previously determined at this temperature shall be maintained for one minute. The voltages of the main and cartridge batteries shall be measured and recorded at one minute intervals during the discharge period. Between discharges and upon completion of the test, the battery shall be recharged.

e. Upon termination of this test, the battery assembly shall be examined for physical damage and leakage of electrolyte.

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4.4.3.4 Shipboard Vibration. The battery assembly shall be mounted in a device which simulates normal target mounting. The battery assembly shall be dry and the battery heater shall not be energized. The battery assembly shall be vibrated at intervals of 1 cps over the range from 5 to 25 cps at a double amplitude of 0.14 inches at 5 to 15 cps and 0.12 inches at 16 to 25 cps along each of three mutually perpendicular axes. At each frequency the vibration shall be maintained for 5 minutes. Resonant frequencies shall be noted. Upon completion of this procedure, the battery assembly shall be vibrated for a total period of two hours at the worst resonant frequency noted along each axis. If no resonant frequency was noted, this portion of the test shall be performed at a double amplitude of 0.14 inches and at a frequency of 25 cps. Upon completion of this test, the battery assembly shall be visually examined for damage and then filled with electrolyte. The battery shall show no leakage of electrolyte.

4.4.3.5 Shock. The battery, with electrolyte added and heater energized, shall be subjected to Procedure V of MIL-E-5272 except that two shocks shall be applied in each of the six directions. During the application of each of the shocks, a simulated target load of 15 amperes shall be placed on the main battery and the cartridge battery. During the discharge period, the voltages of the main and cartridge batteries shall be measured at one minutes intervals and recorded. Upon completion of the test, the battery shall be examined for physical damage and leakage of electrolyte.

4.4.3.6 Fungus. The battery, with electrolyte added, shall be subjected to Procedure I of the fungus test of MIL-E-5272 and shall be examined for damage and leakage of electrolyte. Immediately after the fungus test, the battery assembly shall be subjected to the procedure described in 4.5.4.10.

4.4.3.7 Operating Life. The main and cartridge batteries shall be activated and allowed to soak for not less then one hour, then discharged under a 15 ampere load until the voltages reach 20.9 Vdc for the main battery and 6.6 Vdc for the cartridge battery. Within 24 hours the main and cartridge batteries shall be recharged in accordance with the approved instructions (see 3.1.4). This procedure, less the initial activation and soaking, shall be repeated until five discharges have taken place. The battery shall be examined for damage and leakage of electrolyte.

* 4.5 Production Lot Sample Inspection. The contractor shall supply all quality conformance production lot samples in accordance with the contract or purchase order. Test results shall be made available to the Government. Lot sampling for quality conformance inspection shall be in accordance with MIL-STD-105, Inspection Level II Table I, Single Sampling Plans for Normal Inspection (Master Table) Table IIA. Acceptance Quality Level (AQL) is 4.0 for lot size to be specified in the procurement document. Any corrective action shall be in accordance with the requirements of the contract or puchase order.

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4.5.1 Production Lot Sample Examinations and Tests. Each sample specified in 4.5 shall be subjected to the following examinations and tests.

4.5.1.1 Examinations.

		Requirement
a.	Components	3.1
b.	Instructions	3.1.4
c.	Materials	3.2.2
d.	Moisture, Fungus, and Corrosion	3.2.3
e.	Protective Coatings and Surface Treatments	3.2.4
f.	Battery Marking	3.2.5
g.	Threads	3.3.3
h.	Wiring	3.3.4
i.	Unit packaging	5.1.1
j.	Intermediate Packaging	5.1.2
k.	Packing	5.2
l.	Container Marking	5.3

4.5.1.2 Tests.

		Requirement	Test Method
a.	Dimensions	3.3.1	4.5.4.1
b.	Interchangeability	3.3.2	4.5.4.2
c.	Weight	3.5.1	4.5.4.3
d.	Pressure Tightness	3.5.2	4.5.4.4
e.	Heater	3.5.3	4.4.3.2a
f.	Thermostatic Switch	3.5.4	4.5.4.5
g.	Harness, Connector Receptacle, and heater Resistance	3.5.5.1	4.5.4.6.1
h.	Cell Resistance, Dry	3.5.5.2	4.5.4.6.2
i.	Cell Resistance, Wet	3.5.5.3	4.5.4.6.3
j.	Inversion	3.5.6	4.5.4.7
k.	Cell Voltage	3.5.7	4.5.4.6.3
l.	Capacity, Main Battery	3.5.8	4.5.4.8
m.	Capacity, Cartridge Battery	3.5.9	4.5.4.9
n.	Capacity after 30-Day Stand	3.5.11	4.5.4.10

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4.5.2 Test Conditions. Tests of the battery shall be conducted under the following conditions unless otherwise specified.

a.	Temperature	+65 degrees F to +95 degrees F
b.	Humidity	Up to 95% (relative)
c.	Altitude	Normal Ground
d.	Vibration	None

4.5.3 Test Equipment and Set-Up. The designated testing activity shall furnish and maintain all necessary test equipment, facilities, and personnel to perform all quality conformance tests. The test equipment and facilities shall be adequate in quantity and, unless definite requirements are specified, shall be of sufficient accuracy and quality to permit performance of the required quality conformance production lot sample tests.

4.5.4 Quality Conformance Test Procedure. The quality conformance tests shall be conducted as directed in the following paragraphs. The sequence of tests shall be at the discretion of the testing activity.

4.5.4.1 Dimensions. The battery assembly shall be measured to demonstrate that it conforms to the dimensions and tolerances shown on Drawing 67A76F1.

4.5.4.2 Interchangeability. The following parts and assemblies shall be tested to demonstrate that they are interchangeable within any box assembly without modification or applying force.

a. Main Battery Cells
b. Cartridge Battery Cells
c. Heater, Harness, Receptacle Assembly
d. Thermostatic Switch

4.5.4.3 Weight. The battery, with all cells filled with the required amount of electrolyte, shall be weighted in accordance with U.S. Government standards of weights.

4.5.4.4 Pressure Tightness. Three cells of each sample shall be subjected to an internal pressure of 2.0 plus or minus 0.1 psig. Then the air supply valve shall be closed. At the end of one hour, the pressure shall be measured and shall meet the requirements of 3.5.2.

4.5.4.5 Thermostatic Switch. Prior to installation into the battery assembly, the thermostatic switch shall be subjected to a dielectric test at 500 volts alternating current (Vac) for 60 seconds in accordance with Method 301 of MIL-STD-202. The switch shall then be placed in a test chamber in which the temperature shall be varied through the switch's operating range. An ohmmeter should be placed across the switch terminals to determine whether or not the switch functions. Note and record temperatures at which switch cuts in and out.

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4.5.4.6 Resistance.

4.5.4.6.1 Harness, Connector Receptacle, and Heater. Prior to installation into the battery assembly, the harness, connector receptacle, and heater assembly shall be tested for insulation resistance between each terminal and every other terminal and between each terminal and ground in accordance with Condition B of Method 302 of MIL-STD-202. The harness, connector receptacle, and heater assembly shall be subjected to a dielectric test of 500 Vac for one minute between each terminal and ground in accordance with Method 301 of MIL-STD-202. During and after application of voltage, examine for failure or damage.

* 4.5.4.6.2 Cell, Dry. The internal resistance of three randomly selected individual cells of each sample battery unit in the dry condition shall be measured using test equipment and methods approved by the Government inspector. Each cell shall meet the requirements of 3.5.5.2.

4.5.4.6.3 Cell, Wet. Three cells of each sample battery unit shall be filled with electrolyte and, after soaking for one hour, shall be discharged at a 15 ampere rate until a voltage of 1.41 Vdc per cell has been reached. The cells shall then be recharged. Not less than one hour after completion of charge, the open circuit voltage of each cell shall be measured. The cells shall then be discharged at a 15 ampere rate for 12.5 plus or minus 0.5 minutes and while the cells are still discharging, the voltage shall be read. The resistance of the cells, in a wet condition, shall be calculated as follows:

$$r = \frac{1.60 - E_d}{15}$$

Where: E_d is the discharge voltage of the cell.
r is the wet resistance of the cell.

4.5.4.7 Inversion. One hour after filling, the battery shall be inverted and held in any position. At the end of 30 minutes the battery assembly shall be examined for loss of electrolyte and shall meet the requirements of 3.5.6.

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4.5.4.8 Capacity, Main Battery. Each of two thirds of the main batteries constituting the sample shall be activated (electrolyte added) and allowed to soak for one hour, during which time the battery heater shall be energized from an external power source. The external power source shall then be removed and the main battery, with heater not operating, shall be discharged into a variable resistor at a 15 ampere rate until the main battery voltage has reached 20.9 Vdc. The main battery voltage shall be measured and recorded continuously for the first minute of discharge with a recorder capable of recording voltage during the first five milliseconds of discharge. The battery voltage shall then be measured and recorded at one minute intervals during battery discharge. The battery shall then be recharged.

4.5.4.9 Capacity, Cartridge Battery. Each of two thirds of the cartridge batteries constituting the sample shall be filled with electrolyte and allowed to soak for one hour, during which time the battery heater shall be energized from an external source. The external power source shall then be removed and the cartridge battery, with heater not operating, shall be discharged into a variable resistor at a 15 ampere rate until the cartridge battery voltage has reached 6.6 Vdc. The battery voltage shall be measured at one minute intervals. The cartridge battery shall then be recharged.

4.5.4.10 Capacity, After 30-Day Stand. Each of the remaining one-third of the main and cartridge batteries constituting the sample shall be filled with electrolyte and allowed to stand for 30 days. During the last two hours of the stand period, the battery heater shall be energized from an external power source. The battery heater power shall then be turned OFF, and each battery shall be discharged at a 15 ampere rate into variable resistors until voltages of 20.9 Vdc for the main battery and 6.6 Vdc for the cartridge battery are reached. The main battery voltage shall be measured and recorded continuously for the first minute of discharge with a recorder capable of recording voltage during the first five milliseconds of discharge, and the voltages of both the main and cartridge batteries shall be measured and recorded at one minute intervals during the battery discharge.

4.6 Shipping Condition. The battery shall be packaged and packed as specified in Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging.

5.1.1 Unit Packaging.

5.1.1.1 Battery Assembly. When completed, the battery unit shall be encased in its box assembly (see 3.1.1.2). No additional unit packaging is necessary.

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5.1.1.2 Battery Accessories.

5.1.1.2.1 Installation Screws. The installation screws shall be encased and seal in a close-fitting bag of flexible material conforming to MIL-F-22191, Type II and then placed in the filling kit.

5.1.1.2.2 Filling Kit. The filling kit shall be packaged in accordance with Method IC-2 of MIL-P-116.

* 5.1.1.2.3 Electrolyte. The electrolyte shall be packaged in bottles made of polyethylene conforming to DOT-2E (49 CFR 178.24a). Each bottle shall contain the proper amount of electrolyte to fill one cell. Each bottle shall be marked to indicate FULL and the electrolyte level shall be visible in relation to the mark.

5.1.1.3 Instructions. Each set of instructions shall be encased and sealed in a close-fitting bag of flexible material conforming to MIL-F-22191, Type II.

5.1.2 Intermediate Packaging.

5.1.2.1 Battery Assembly. The battery assembly shall be encased and sealed as specified on Drawing 2580161.

5.1.2.2 Electrolyte. Each bottle of electrolyte shall be encased and sealed as specified on Drawing 2580161.

5.1.2.3 Filling Kit. The filling kit shall be encased and sealed as specified on Drawing 2580161.

5.2 Packing.

5.2.1 Battery. The battery, with its set of instructions, shall be packed in accordance with Drawing 2580161.

5.3 Marking. Marking of unit packages and shipping containers shall be in accordance with MIL-STD-129, and shall include serial number, date of manufacture, arrows (to indicate "up" position of container), "Property of U.S. Government", and a hazardous chemical notation to read: "CAUTION CONTAINS HARMFUL CAUSTIC SOLUTION. AVOID INHALATION OF ELECTROLYTE FUMES AND EXTERNAL CONTACT WITH BODY OR CLOTHING". Markings, except for the arrows indicating the container shipping position, shall be provided on a label.

6. NOTES

6.1 Intended Use. The battery is intended to be used as the primary source of electrical power in the AQM-37A Missile Target.

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6.2 Ordering Data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. The number of battery assemblies to be provided.
c. All options or deviations.
d. Activity designated by procuring activity.

* 6.2.1 Data Requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Armed Services Procurement Regulations, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR-7-104.9(n) are not invoked, the data will be delivered by the contractor in accordance with the contractor requirements.

Paragraph	Data Requirements	Applicable DID
4.4	Report, First Article Test	UDI-T-21349

"(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer)."

* 6.3 Superior Substitutes. Recommendations concerning the use of a superior substitute for any part or material specified herein shall be accompanied by a description of the proposed substitute and the reasons therefore, together with evidence to substantiate any claim as to its suitability, and shall be submitted to the procuring activity for approval. At the discretion of the procuring activity, test samples may be required to prove the suitability of the proposed substitute.

* 6.4 Precedence of Documents. When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract. The contract shall have precedence over any specification.

(2) This Specification. This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications. Any referenced specification shall have precedence over all applicable subsidiary specifications referenced herein. All referenced specifications shall apply to the extent specified.

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6.5 Changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire relationships to the last previous issue.

* 6.6 Approval for Shipment. Battery packaging in accordance with Drawing 2580161 is approved for surface shipment under 49 CFR 173.7a by NAVSEASYSCOM letter 06G3/GSM/4030 SER 214 of 27 October 1978.

* 6.7 First Article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. The first article should be a preproduction sample. The first article should consist of five units. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examinations, test and approval of the first article.

Preparing Activity:
Navy-AS

Project 6140-N519

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL	OMB Approval No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-B-81502(AS) BATTERY, SILVER-ZINC-ALKALI

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO (If "Yes", in what way?)

4. REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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